

SYSTEM ARCHITECTURE NOTE: DATABASE, AUTH & EMAIL SYSTEM

1. DATA MODELS & RELATIONSHIPS

The application uses SQLite with Prisma ORM to store data across five models: Role, User, Transaction, AuditLog, and EmailLog. I configured foreign keys and delete rules so related records stay organized and clean:

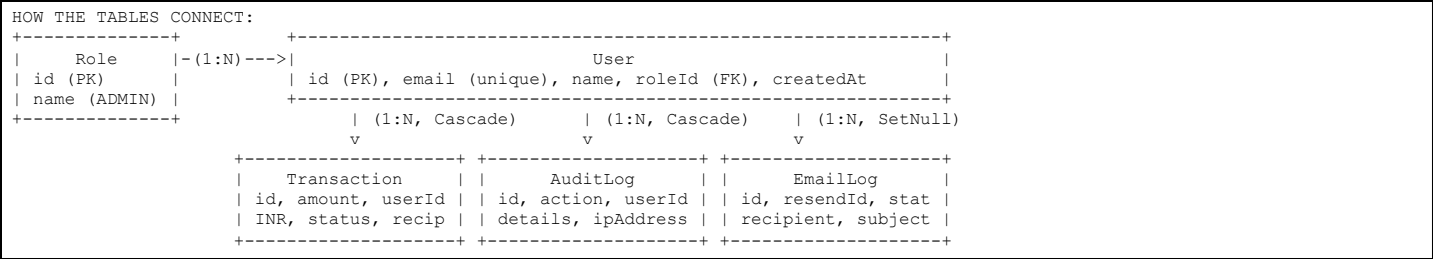


TABLE NAME	PRIMARY KEY	FOREIGN KEYS	RELATIONSHIP & DELETE BEHAVIOR
Role	id (CUID)	name (Unique)	One role has many users. A role cannot be deleted if users are still assigned to it (Restrict).
User	id (CUID)	email, roleId	Each user has one role, and can own multiple transactions, audit logs, and emails.
Transaction	id (CUID)	userId	Belongs to a user. If the user account is deleted, their transactions are deleted too (Cascade).
AuditLog	id (CUID)	userId	Tracks user actions (like creating a transaction). Deleted automatically if the user is deleted (Cascade).
EmailLog	id (CUID)	resendId, userId	Stores email dispatch history. If a user is deleted, email logs are kept with user set to null (SetNull).

2. TERMINAL LOGS: MIGRATION & SEED EXECUTION

I updated the SQLite database schema using `npx prisma db push`, then ran `npx prisma db seed` to populate the database with real users and test transactions in Indian Rupees (₹):

LOG 1 - TERMINAL: PRISMA MIGRATION (NPX PRISMA DB PUSH)

```
roystonsoans@Roystons-MacBook-Air-3 FST1 % npx prisma db push
warn The configuration property 'package.json#prisma' is deprecated and will be removed in Prisma 7. Please migrate to a Prisma config file (e.g., 'prisma.config.ts'). For more information, see: https://pris.ly/prisma-config
Environment variables loaded from .env
Prisma schema loaded from prisma/schema.prisma
Datasource "db": SQLite database "dev.db" at "file://dev.db"

The database is already in sync with the Prisma schema.
✓ Generated Prisma Client (v6.19.3) to ./node_modules/@prisma/client in 41ms
```

LOG 2 - TERMINAL: DATABASE SEED (NPX PRISMA DB SEED)

```
roystonsoans@Roystons-MacBook-Air-3 FST1 % npx prisma db seed
warn The configuration property 'package.json#prisma' is deprecated and will be removed in Prisma 7. Please migrate to a Prisma config file (e.g., 'prisma.config.ts'). For more information, see: https://pris.ly/prisma-config
Environment variables loaded from .env
Running seed command 'tsx prisma/seed.ts' ...
✓ The seed command has been executed.
```

3. DATABASE MANAGEMENT TOOL: VIEWING SAVED RECORDS

I opened Prisma Studio (running at `http://localhost:5555`) to inspect the data stored in the database:

LOG 3 - DATABASE TOOL: SAVED USERS & ASSIGNED ROLES (PRISMA STUDIO)

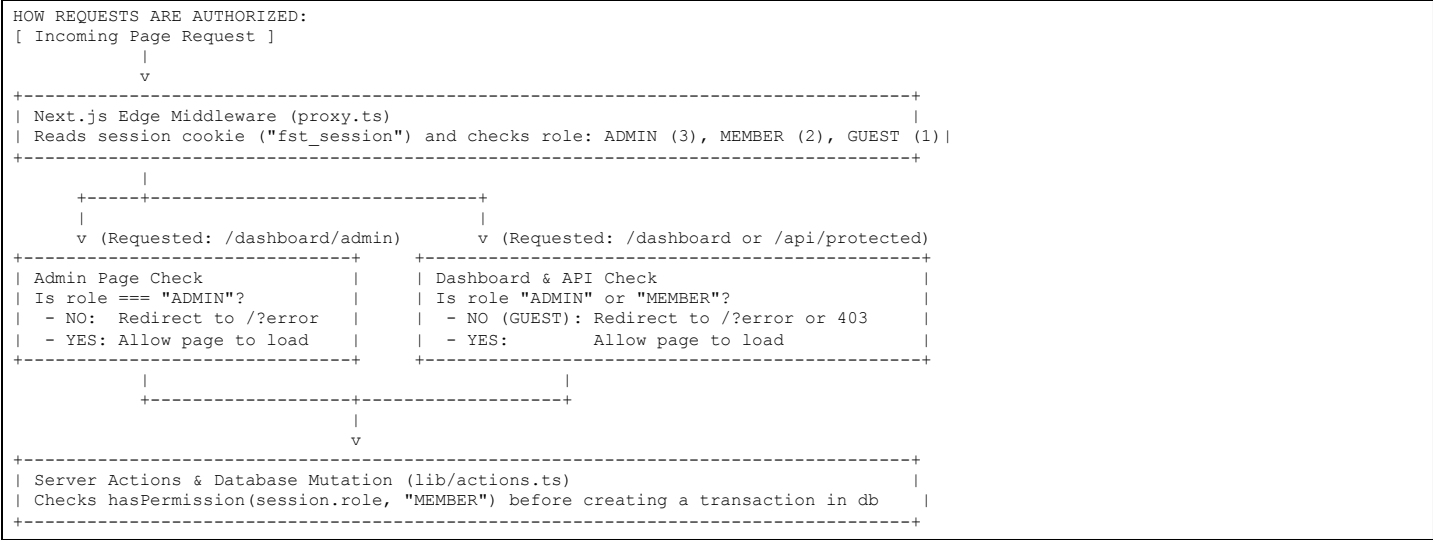
User X Transaction X Role X AuditLog X Close all										
All Models	Filters	None	Fields	All	Showing	3 of 3	Add record			
AuditLog	3		id A	email A	name A	roleId A	role ()	transactions ()	auditLogs ()	email
EmailLog	2		cmua43rks000a1ox37fjb...	roystonsoans3@gmail.c...	Royston Soans	cmua43rko0001ox33cof...	Role	4 Transaction	3 AuditLog	2 E
Role	3		cmua43rkr00061ox3ypo8...	member@ledgercraft.dev	Aarav Mehta	cmua43rkp00011ox30kvg...	Role	2 Transaction	0 AuditLog	0 E
Transaction	6		cmua43rku00081ox3s96p...	guest@ledgercraft.dev	Ananya Sharma	cmua43rkq00021ox3omz6...	Role	0 Transaction	0 AuditLog	0 E
User	3									

LOG 4 - DATABASE TOOL: SAVED LEDGER TRANSACTIONS IN INR (PRISMA STUDIO)

Transaction X Role X AuditLog X Close all										
All Models	Filters	None	Fields	All	Showing	6 of 6	Add record			
AuditLog	3		id A	amount #	currency A	status A	recipient A	category A	reference A?	user
EmailLog	2		cmua43rks000a1ox37ojm...	14500	INR	COMPLETED	Amazon Web Services L...	Cloud Infrastructure	REF-AWS-9042	cmua
Role	3		cmua43rkt000c1ox3xskx...	6200	INR	COMPLETED	Google Cloud Platform	API Billing	REF-GCP-8114	cmua
Transaction	6		cmua43rku000e1ox36k1o...	75000	INR	COMPLETED	Engineering Stipend	Payroll	REF-PAY-3301	cmua
User	3		cmua43rkv000g1ox3kpc...	3800	INR	COMPLETED	GitHub Enterprise	Security	REF-GH-5520	cmua
			cmua43rkv000i1ox3fi03...	9450	INR	PENDING	Apple India Retail	Hardware	REF-APL-7741	cmua
			cmua43rkv000k1ox3us4m...	4100	INR	COMPLETED	Slack Technologies	SaaS Licensing	REF-SLK-1290	cmua

4. ROLE-BASED ACCESS CONTROL (RBAC) FLOW

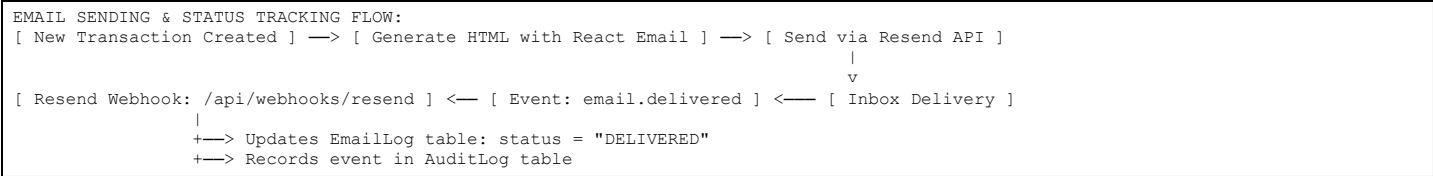
I protected pages and actions using two layers of checks: 1. **Edge Middleware (proxy.ts)**: Runs before a page loads. It reads the session cookie (fst_session) to block or redirect unauthorized visitors right away. 2. **Server Actions (lib/actions.ts)**: Runs on the server when form data is submitted, checking user permissions before updating the database.



USER ROLE	ACCESS LEVEL	HOME PAGE (/)	DASHBOARD (/DASHBOARD)	ADMIN PAGE (/DASHBOARD/ADMIN)	CREATE TRANSACTION
ADMIN	Level 3 (Full)	Full Access	Full Access	Full Access (Can see all users)	Allowed
MEMBER	Level 2 (Standard)	Full Access	Full Access	Blocked (Redirects to home)	Allowed
GUEST	Level 1 (View-Only)	View Only	Blocked (Redirects to home)	Blocked (Redirects to home)	Blocked (Error shown)

5. TRANSACTIONAL EMAIL SYSTEM & WEBHOOK TRACKING

When a user creates a transaction, the app sends an email confirmation using React Email and the Resend API, and tracks delivery status through webhooks:



LOG 5 - DATABASE TOOL: PERSISTED EMAIL DISPATCH LOGS (PRISMA STUDIO)

	EmailLog X	User X	Transaction X	Role X	AuditLog X	Close all	
	Filters	None	Fields	All	Showing 2 of 2	Add record	
	id A	recipient A	subject A	status A	resendid A?	userid A?	user ()?
	cmua43rkq00s1ox3u4t7...	roystonsoans3@gmail.c...	Transaction Confirmat...	DELIVERED	01a0bf7f-efad-777c-a8...	cmua43rkq00041ox3vfsb...	User 2026
	cmua43rkx000u1ox3yy91...	roystonsoans3@gmail.c...	Monthly Statement & S...	DELIVERED	01a0bf7f-b88a-777c-b2...	cmua43rkq00041ox3vfsb...	User 2026

5.2 Webhook Handling & Bounce Management

When Resend sends an update to the webhook endpoint (/api/webhooks/resend), the server checks the message ID and updates the record in the EmailLog table. If an email fails to deliver, its status is changed to BOUNCED, and an entry is added to the AuditLog table so I can see what happened.